

What if I told you that hackers have a 'standard operating procedure' just like a Fortune 500 company? Every major breach—from Facebook to the FBI—follows a specific, predictable blueprint.

Today, I'm giving you the Master Key to the attacker's playbook. Master these two frameworks, and you stop being the prey and start being the hunter.  

What is the Cybersecurity Kill Chain?

- A framework describing stages of a cyber attack
- Developed by **Lockheed Martin**
- Shows how attackers progress from
recon → objectives
- Helps defenders detect and stop attacks early
- Linear, phase-based security model



Why It Matters in Defense

- Identifies where to detect attackers early
- Improves incident response planning
- Enables proactive security posture
- Helps build layered defense strategy
- Supports threat hunting and SOC monitoring

Stages of the Cyber Kill Chain

- **Reconnaissance** — attacker gathers target info
- **Weaponization** — payload + exploit prepared
- **Delivery** — phishing, USB, web, etc.
- **Exploitation** — vulnerability triggered
- **Installation** — malware persistence established
- **Command & Control (C2)** — remote attacker communication
- **Actions on Objectives** — data theft, damage, etc.



The Concept of "Chokepoints"

- Points where defenders can stop the attack
- Earlier disruption = lower damage
- Email gateways, EDR, IDS act as chokepoints
- Focus on detect → deny → disrupt → degrade
- Goal: break the attacker's chain

Real-World Attack **(Kill Chain Mapping)**

- Recon → collect employee emails
- Delivery → send malicious attachment
- Exploitation → user opens file
- Installation → malware persists
- C2 → attacker gains remote control
- Objective → credential theft / data exfiltration
- Shows defenders where detection is possible

Limitations of the Traditional Kill Chain



What is MITRE ATT&CK Framework?

- A knowledge base of real-world attacker behavior
- Maintained by MITRE Corporation
- Maps adversary TTPs (Tactics, Techniques, Procedures)
- Used by red team, blue team, SOC, threat hunters
- Matrix-based, not linear

Structure of MITRE ATT&CK

- Tactics = attacker goals (WHY)
- Techniques = methods used (HOW)
- Procedures = real-world implementations
- Organized as ATT&CK Matrix
- Covers Enterprise, Mobile, and ICS environments

MITRE ATT&CK **Tactics Overview**

-  Reconnaissance
-  Resource Development
-  Initial Access
-  Execution
-  Persistence
-  Privilege Escalation
-  Defense Evasion
-  Credential Access
-  Discovery
-  Lateral Movement
-  Command & Control
-  Exfiltration
-  Impact



Kill Chain vs MITRE ATT&CK

- Kill Chain = linear attack flow
- MITRE = detailed behavior matrix
- Kill Chain = high-level strategy
- MITRE = granular detection mapping
- Best practice: use both together

How Defenders Use These Frameworks



Corporate Security Benefits



Common Misconceptions